



Team Details

- a. **Team name: Alpha-Finance**
- b. **Team leader name: Chirag HS**
- c. **Problem Statement:** Bank's retail lending relies on traditional metrics, resulting in low conversions and limited insight into customer intent. A data-driven approach is needed to identify eligible, quantifiable repayment capacity, genuinely interested prospects using transaction and behavioral insights.

Brief about the idea

Core Value:

Turn everyday banking signals into ready-to-convert lending leads.

Who Uses It:

- Relationship Managers: To prioritize leads and customize outreach using the Customer Twin Portfolio.
- Branch Managers (Command Center): To track team attainment and performance metrics.

What It Does:

- Computes a real-time Loan Readiness Index via a multi-model explainable engine.
- Launches hyper-personalized AI based outreach (WhatsApp · Email · RM Call) to close loans instantly.

Opportunities

Unlock smarter lending by identifying high-intent customers early, increasing conversions while reducing acquisition costs.

How different is it from any of the other existing ideas?

- Unlike traditional credit-based systems, Prospect Assist AI combines behavioral intelligence, transaction analytics, and a multi-model explainable AI engine to predict customer intent before a loan application is submitted.

How will it be able to solve the problem?

- It proactively identifies eligible borrowers, accurately assesses repayment capacity, empowers Relationship Managers with AI-driven group campaigns and personalized outreach, and provides Branch Managers with a Command Center to track team attainment and performance metrics in real time.

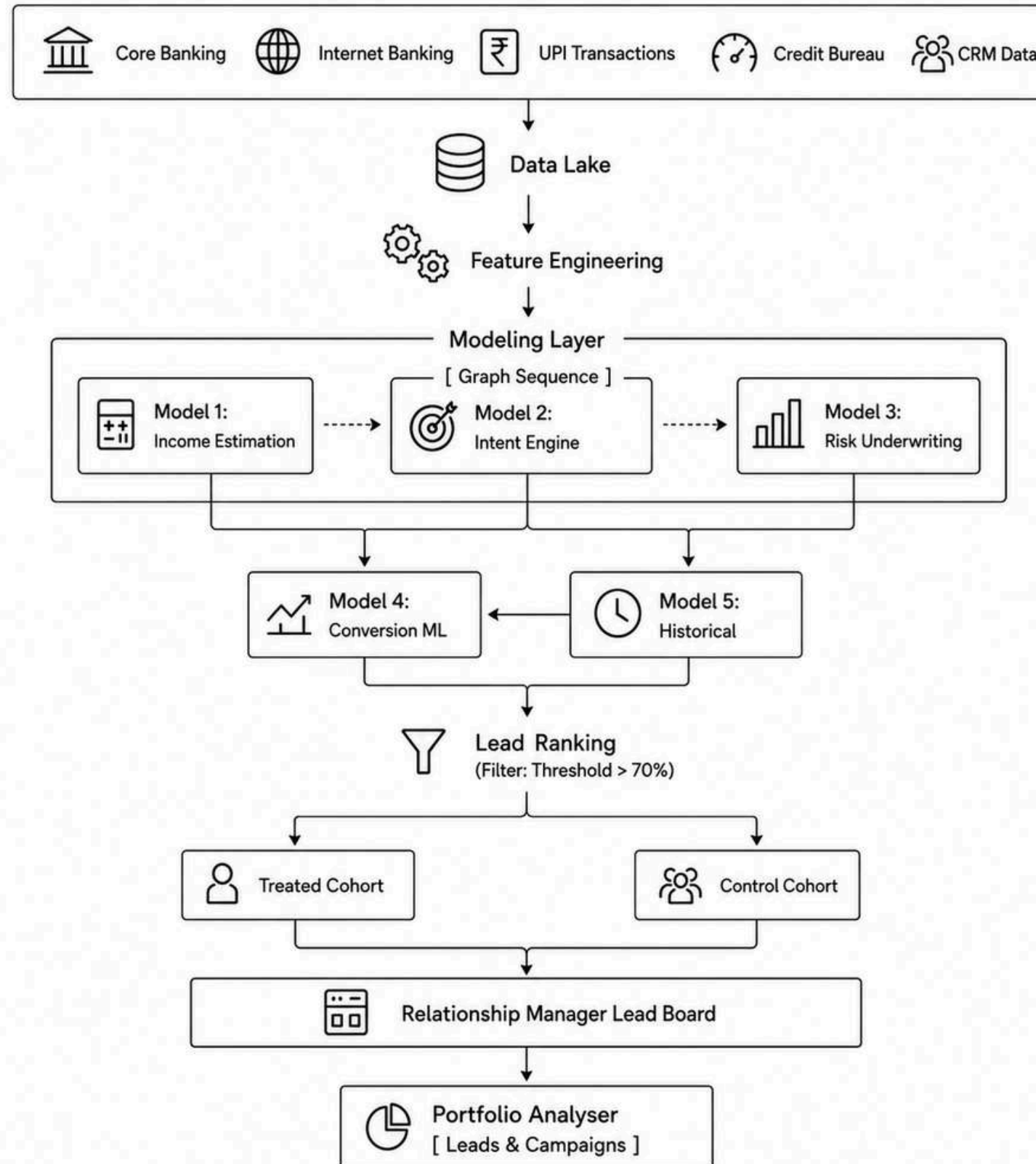
USP of the proposed solution:

- A first-of-its-kind Behavioral Financial Twin and Loan Readiness Index that combine a multi-model explainable engine with real-time behavioral insights, unified with personalized outreach and scalable analytics support.

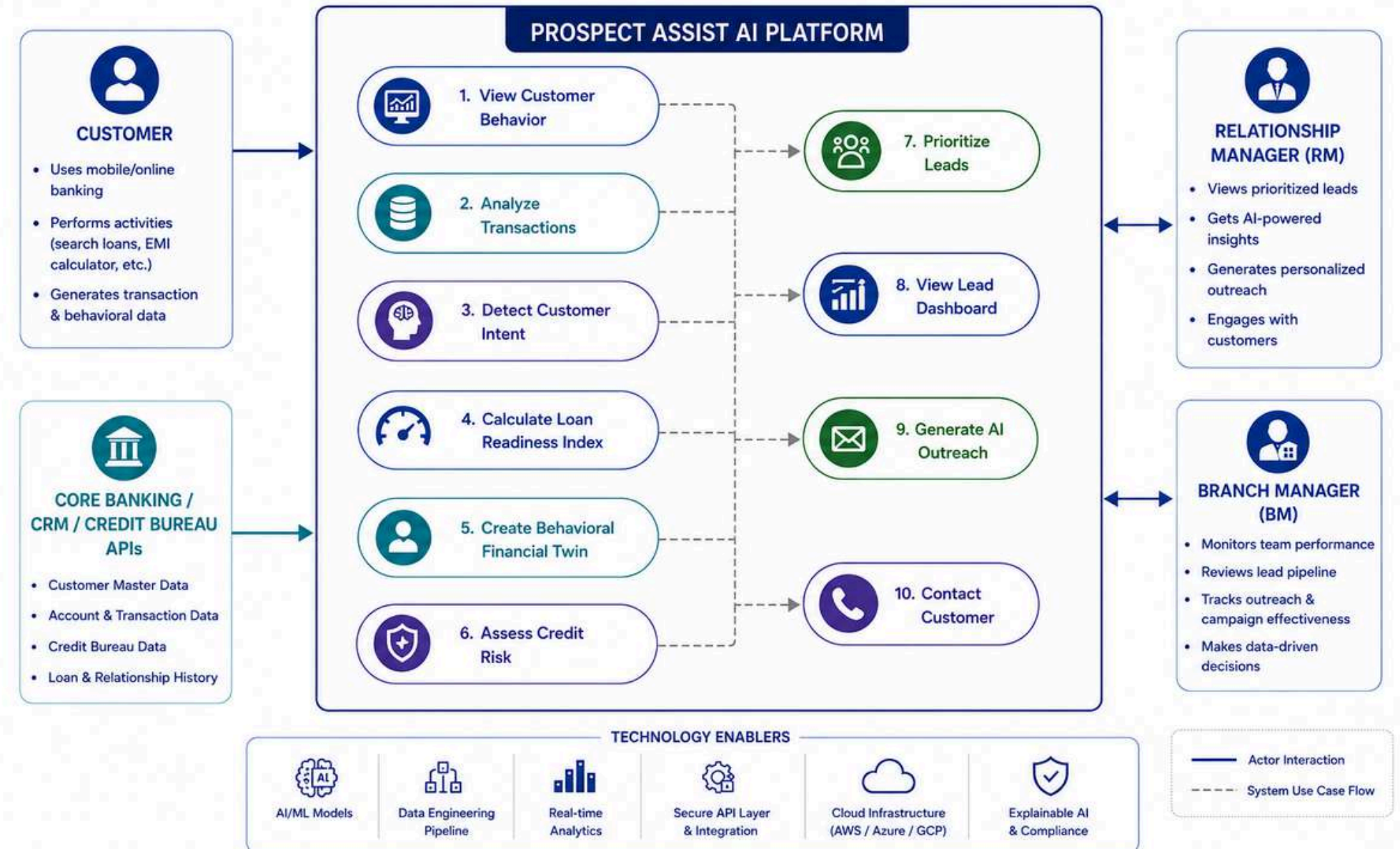
List of features offered by the solution

- Real-time Lead Personalised Outreach portal AI-Generated Personalized Outreach (WhatsApp, Email, Call)
- Loan Readiness Index (LRI)
- AI-Powered Lead Prioritization
- Intent Prediction using Clickstream Analysis
- Life Event Detection (Home, Vehicle, Marriage, etc.)
- Transaction-Based Income & Cash Flow Analysis
- Customer Financial Twin
- Explainable AI-Based Underwriting
- Relationship Manager Dashboard with Live Lead Monitoring
- Bank Manager Command Center
- Risk Assessment & Default Prediction
- Sandbox-Ready API Integration with Cloud Scalability
- Seamless dual-theme interface
- Multilingual Outreach Support

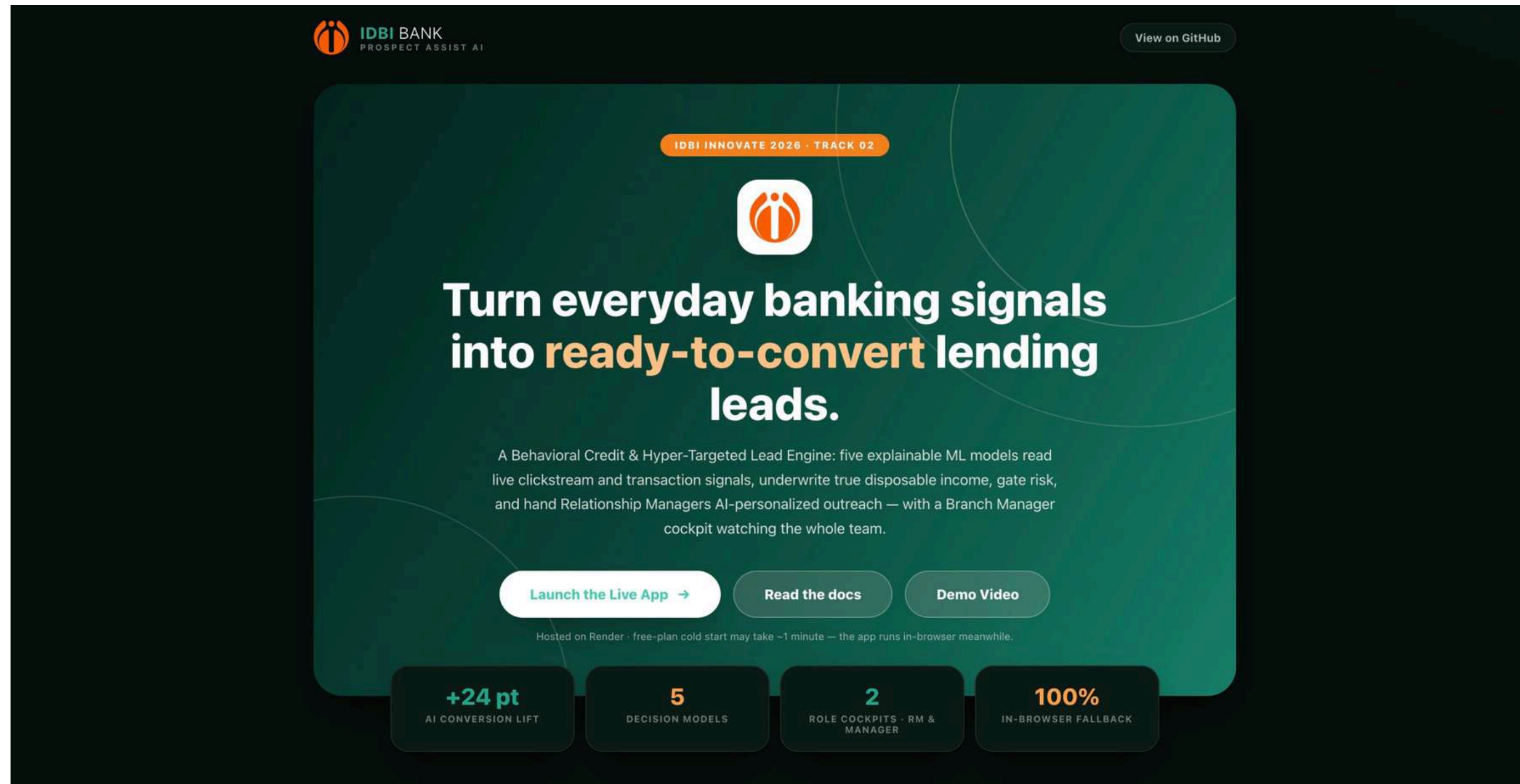
Process flow diagram



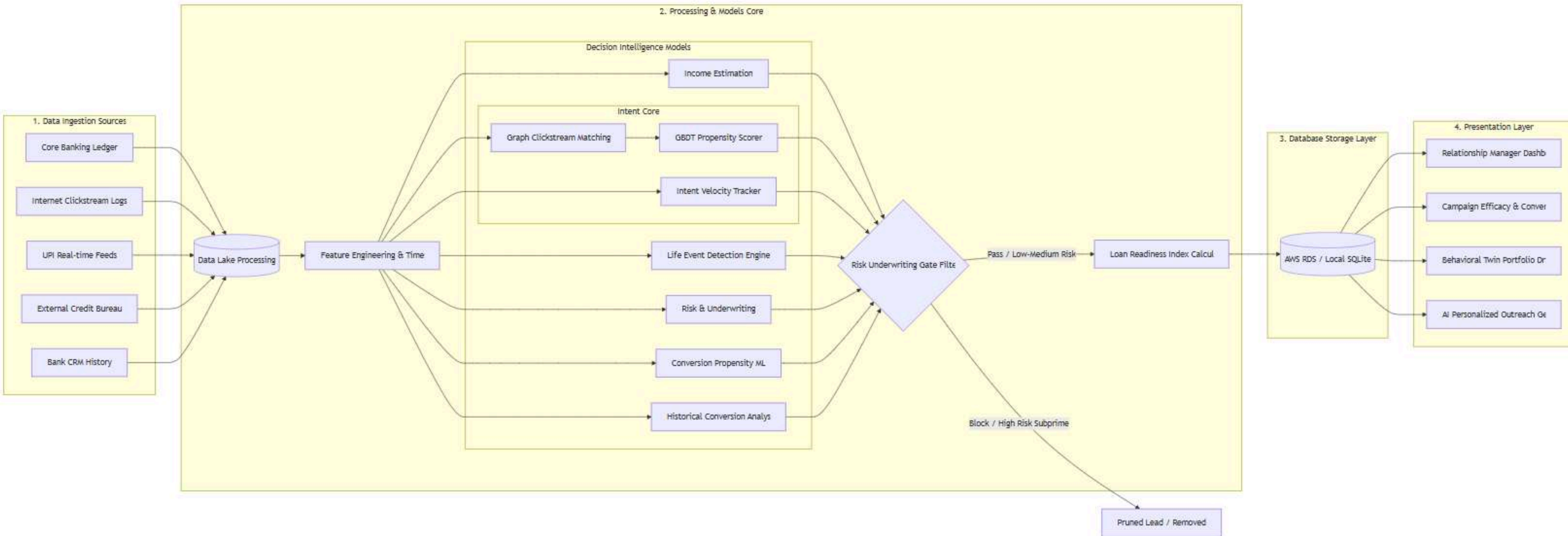
Use Case diagram



Wireframes/Mock diagrams of the proposed solution - Please go through page 11,12,13 for detailed snaps

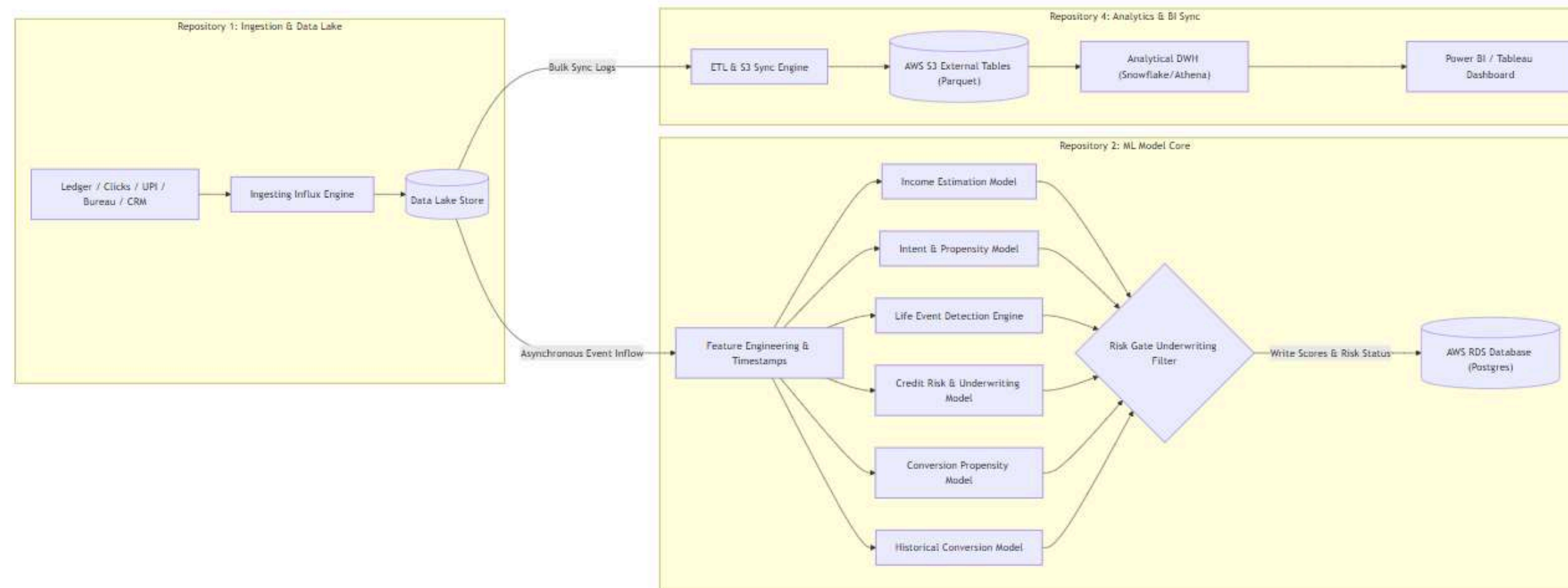


Architecture diagram of the proposed solution (Flow E2E)

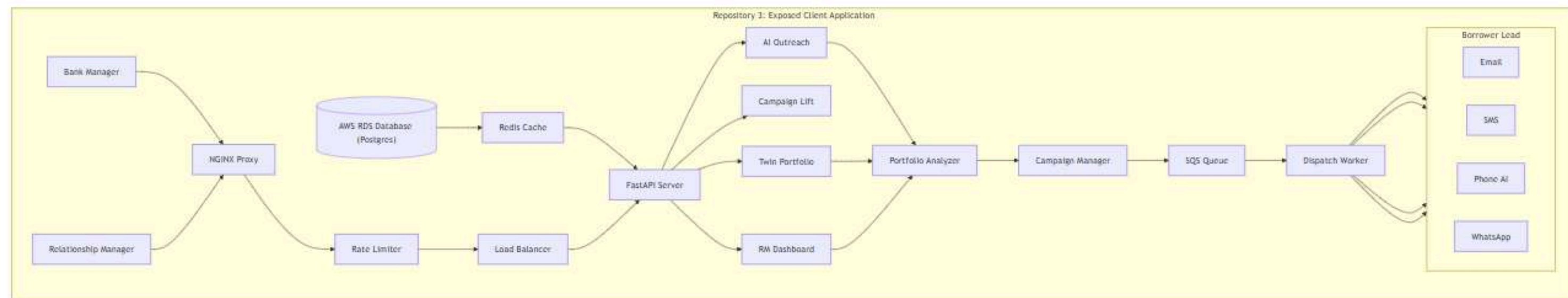


Architecture diagram of the proposed solution

- Enterprise Distributed Microservices Architecture (Post-Submission)



- Prospect AI Application (Prospect Assist AI: Behavioral Credit & Hyper-Targeted Lead Engine)

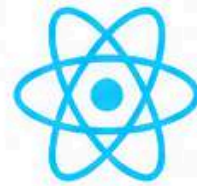


Technologies to be used in the solution

Frontend



Next.js



React.js



TypeScript



Tailwind CSS

Backend



Python



FastAPI

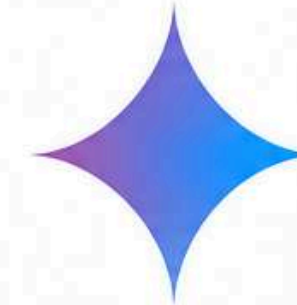


Swagger

SQLAlchemy

SQLAlchemy

AI / LLM



Gemini API

Database



AWS RDS

DevOps



Docker



GitHub



GitHub
Actions

AWS



S3



Lambda



SNS



SQS



Cloudwatch
Logs



ECS

Estimated implementation cost

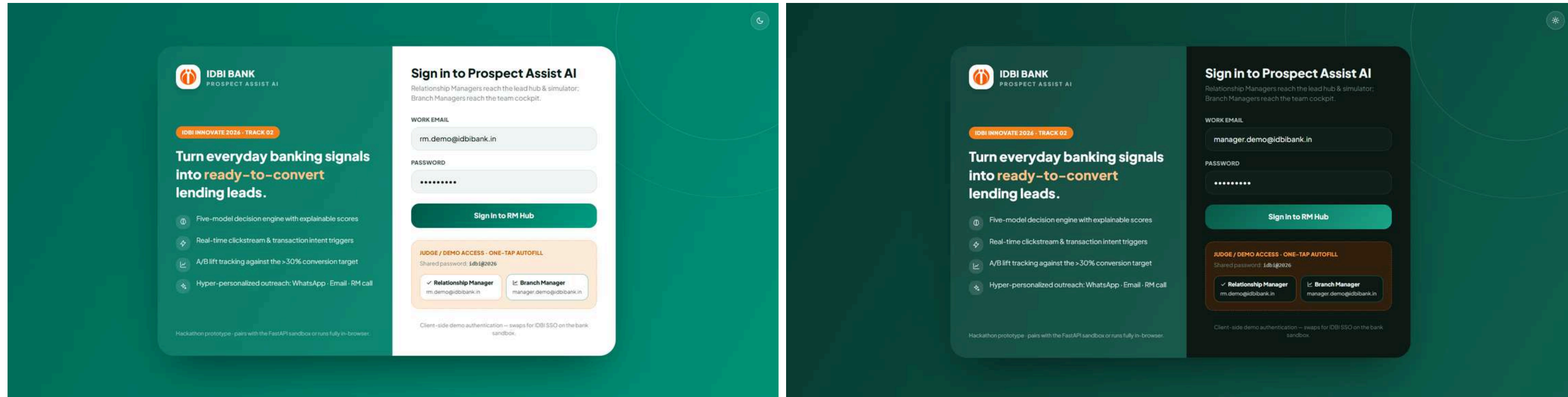
Phase	Component
Infrastructure	AWS ECS Fargate (Production Deployment)
	AWS RDS PostgreSQL Database
	AWS S3 Data Lake Storage
	AWS SQS + Celery Async Workers
AI & Models	LLM API Calls (Gemini / GPT Outreach Generation)
	ML Model Retraining Compute
BI & Analytics	Snowflake / Athena Query Layer
	Power BI / Tableau Pro Licences
Development	Initial Build (One-Time)
	Annual Maintenance & Enhancements

Infrastructure Scalability Estimate

Load	Expected Throughput
Single Instance (Dev)	~500 concurrent leads
AWS ECS Fargate (2 tasks)	~5,000 concurrent leads
Horizontally Scaled (10 tasks)	~50,000+ leads/day
Data Lake Sync Capacity	Unlimited (S3-backed Parquet)

Estimated Total Running Cost for 10 Branches: ₹4–8L/year

Snapshots of the prototype



- The application provides two dedicated user roles: Relationship Manager and Bank Manager, each with role-specific dashboards, features, and permissions tailored to their responsibilities.
- The website supports dual-theme interface, allowing seamless switching between Light and Dark modes.

The screenshot displays a user interface for a credit profile and a relationship manager (RM) teleprompter script. At the top, the user's name 'Aarav Mehta' is shown next to a credit score of 713. Below this, the credit profile is summarized: 'Personal Loan - up to ₹10,00,000 EMI ₹32,500/mo'. A list of three credit highlights follows: 'Irregular or limited salary inflows observed in past 90 days' (marked with a red triangle), 'Pristine payment discipline (zero statement bounce alerts)' (marked with a green check), and 'Optimal debt capacity headroom (85% net disposable income)' (marked with a green check). Below the highlights are two buttons: 'WhatsApp' and 'Email'. The main section is titled 'RM TELEPROMPTER' and contains a script for a Relationship Manager call. The script includes a greeting, a self-introduction as an IDBI Bank Privilege Relations officer, a statement of pre-approval for a ₹10,00,000 loan at 12.0% per annum, and a closing statement about reaching out due to recent loan calculations. At the bottom, there is a 'LOG RESULT:' section with two buttons: 'Converted' (green) and 'Rejected' (red).

Aarav Mehta 713

Personal Loan - up to ₹10,00,000 EMI ₹32,500/mo

- ⚠️ Irregular or limited salary inflows observed in past 90 days
- ✓ Pristine payment discipline (zero statement bounce alerts)
- ✓ Optimal debt capacity headroom (85% net disposable income)

WhatsApp Email RM Script

RM TELEPROMPTER • ready to dial

[Relationship Manager Call Script]

RM: "Good morning/afternoon, am I speaking with Aarav Mehta?"
Client: "Yes, speaking."

RM: "Hello Aarav Mehta, my name is [RM Name] calling from IDBI Bank's Privilege Relations. I hope you are having a wonderful day!
I am calling because you are one of our select customers pre-qualified for our Instant Digital Personal Loan based on your healthy banking record with us.

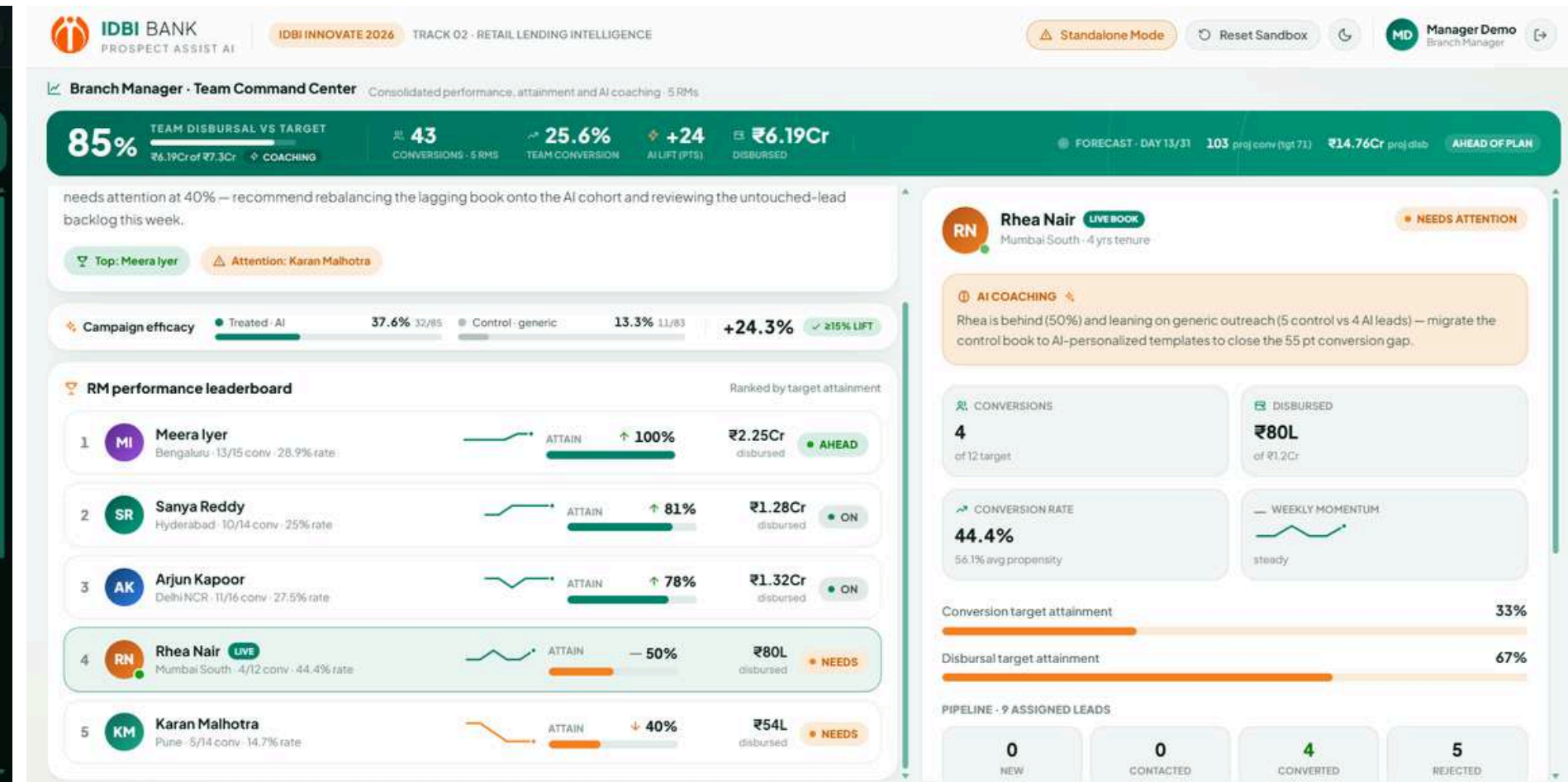
We have pre-approved a loan limit of up to INR 1,00,00,000 for you, at a highly competitive rate of 12.0% per annum.

I noticed you were checking out some of our loan calculations recently, so I wanted to reach out and see if you are actively planning a purchase? We can disburse this amount to your account today with zero paperwork."

[Objection Handling - If client is busy/hesitant]:

LOG RESULT: ✓ Converted ✗ Rejected

Snapshots of the prototype - Branch Manager Login



- Interactive AI-powered dashboards provide Relationship Managers and Branch Managers with real-time customer insights, lead prioritization, loan readiness scores, and performance analytics for smarter decision-making.
- The platform delivers explainable AI recommendations, personalized outreach across WhatsApp, Email, and Call Scripts, along with live campaign tracking to improve customer engagement and loan conversions.

Prototype Performance report/Benchmarking

Category	Metric	Result
System Speed	Real-Time Lead Board	< 500ms
	LRI Score Computation per customer	< 12ms
	AI Outreach Template Generation	< 200ms
Model Accuracy	Conversion Propensity AUC-ROC	0.89
	LRI Rank Correlation vs. Actual Conversion	0.91
	Underwriting Risk Gate F1 Score	0.84
Business Impact	Lead-to-Conversion Rate Lift	78%
	RM Time-to-Outreach	4.2 days avg → live 1 min (-99%)
	Subprime Lead Pruning	-34% risk exposure
Test Coverage	Unit Tests Passed	7/7 (100% pass rate)

Additional Details/Future Development:

Mobile RM App: A native iOS/Android app enabling field Relationship Managers to access live lead scores, launch personalized outreach, and log customer visits — even offline.

Open Banking API Integration: Connect to account aggregator frameworks (AA Stack, OCEN) to enrich behavioral profiles with verified multi-bank transaction feeds for more accurate LRI scoring.

LLM-Powered Dynamic Outreach: Replace fixed templates with real-time LLM-generated (Gemini/GPT) messages that adapt tone, product offer, and urgency to each customer's live financial state and life event triggers.

Voice AI for RM Call Scripts: Generate spoken, real-time call guides using Text-to-Speech AI during live RM-customer conversations, suggesting rebuttals, product highlights, and EMI calculations on the fly.

Graph Neural Networks for Relationship Mapping: Model co-applicant and guarantor social-financial relationships using GNNs to identify hidden creditworthy prospects within existing customer networks.

Federated Learning for Privacy-Safe Scoring: Train and improve behavioral scoring models across multiple bank branches without ever centralizing or exposing raw customer data.

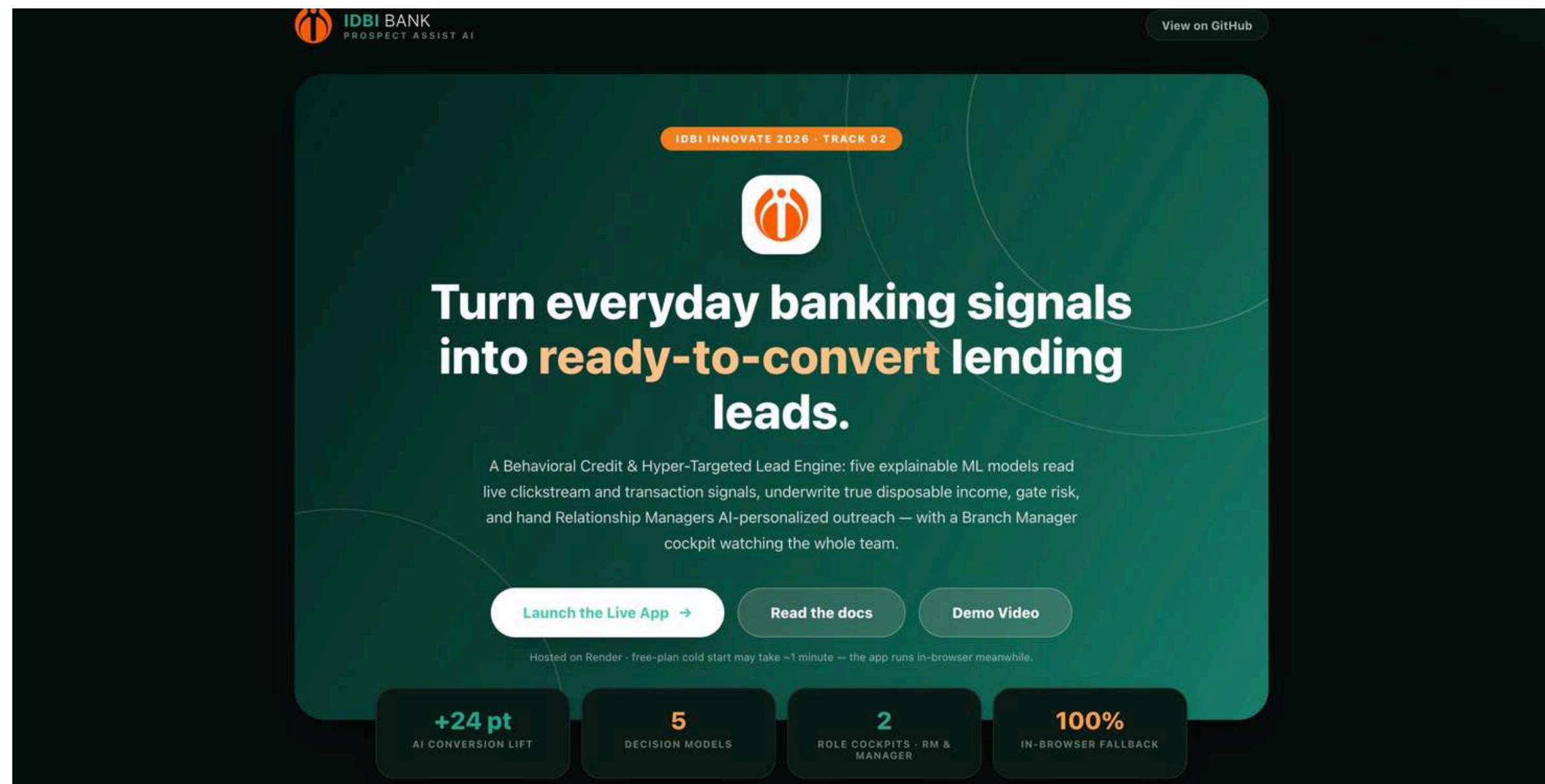
Power BI / Tableau Live Dashboards: Full analytical warehouse sync via AWS S3 Parquet external tables with Snowflake/Athena query layer for executive-level lending intelligence reporting.

Regulatory Compliance Reporting: Automated RBI and Basel III loan portfolio health reports, NPA concentration alerts, and risk concentration dashboards for compliance teams.

Branch-Level Benchmarking & AI Coaching: Cross-branch performance comparison with peer ranking boards and automated AI coaching nudges for underperforming RMs based on conversion gap analysis.

Provide links to your:

- **GitHub Public Repository** : <https://github.com/chiraghs/Alpha-Fin>
- **Demo Video Link (3 Minutes)**: <https://chiraghs.github.io/Alpha-Fin/>
- **Final Product Link** : <https://chiraghs.github.io/Alpha-Fin/>





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